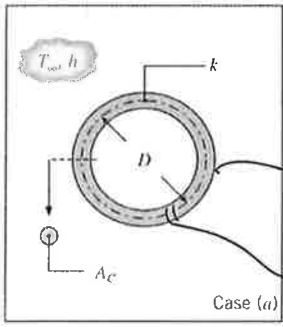
PROBLEM 5.6

100/100

Rayan Baig, CHE 311, Homework # 6

For each of the following cases, determine an appropriate characteristic length (L_c) and corresponding Biot number B_i that is associated w/ the transient thermal response of the solid object. State whether the lumped capacitance method is valid. If the temperature information is not provided, evaluate properties at $T = 300k$.

A) A toroidal shape of diameter $D = 50mm$ and cross-sectional area $A_c = 5mm^2$ is of thermal conductivity $k = 2.3\ W/mk$. The surface of the torus is exposed to a coolant corresponding to a convection coeff. of $h = 50\frac{w}{m^2k}$



Case (a)

Assumptions: * Constant properties

$L_{c1} = \frac{V}{A_s}$,

$$L_{c1} = \frac{A_c^{\frac{1}{2}} \cancel{\pi D}}{2\pi^{\frac{1}{2}}\left(\sqrt{\frac{A_c}{\pi}}\right)\cancel{\pi D}} = \frac{1}{2}\sqrt{\frac{A_c}{\pi}} = \frac{1}{2}\sqrt{\frac{5mm^2 \times}{\pi}\frac{m^2}{(1000mm)^2}}$$

Outside $\pi D \times A =$ Volume

$\star\ L_{c1} = 0.0006308m$

$A_s =$ surface area ∴ increment has radius r_o using

$A_c = \pi r_o^2 \rightarrow$

$r_o = \sqrt{\frac{A_c}{\pi}}$

Now find circumference

$\underbrace{2\pi r_o \times \pi D}_{A_s\ ∴}$

Now finding the max center to surface radius

$\text{radius}_{max} = L_{c2} = \sqrt{\frac{A_c}{\pi}} = 0.001262m$

25/25

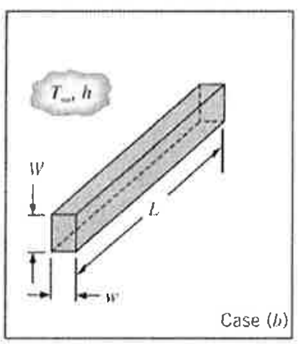
Testing B_i for Both characteristic lengths

$$\#1(L_{c1}) \Rightarrow B_i = \frac{hL_{c1}}{k} = \frac{\left(50\frac{w}{m^2k}\right)(0.0006308m)}{2.3W/mk} = \underline{0.0137 < 0.1}$$

$$\#2(L_{c2}) \Rightarrow B_i = \frac{hL_{c2}}{k} = \frac{(50\frac{w}{m^2k})(0.001262m)}{2.3\frac{w}{mk}} = \underline{0.0274 < 0.1}$$

Lumped Capacitance approximation is valid according to both definitions.

B) A long, hot AISI 304 stainless steel bar of rectangular cross section has dimensions $w = 3mm$, $W = 5mm$, and $L = 100mm$. The bar is subject to a coolant that provides a heat transfer coefficient of $h = 15w/m^2k$ @ all exposed surfaces.



Case (b)

$$L_{c1} = \frac{V}{A_s} = \frac{WwL}{2((wW)+(WL+wL))} = \frac{(5mm)(3mm)(100mm)}{2(5\times3 + 100(5+3))\ mm^2}$$

$$= 0.00092m$$

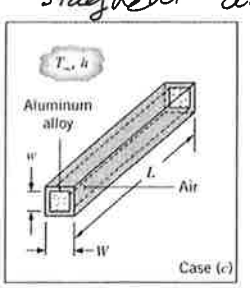
For (L_{c1})

$$B_i = \frac{hL_{c1}}{k} = \frac{(15\frac{w}{m^2k})(0.00092m)}{14.9\ W/m \cdot k} = \underline{0.000926 < 0.1}$$

Tab. →

Lumped Capacitance approx is valid.

C) A long extruded aluminum (Alloy 2024) tube of inner and outer dimensions $w = 20mm$ & $W = 24mm$ respectively, is suddenly submerged in water, resulting in a convection coeff. of $h = 37W/m^2k$ @ the four exterior tube surfaces. The tube is plugged @ both ends trapping stagnant air inside the tube.



Case (c)

$$L_{c1} = \frac{V}{A_s} = \frac{\cancel{L}(W^2 - w^2)}{4(\cancel{L} \times W)} = \left(\frac{(24^2 - 20^2)mm^2}{24mm}\right) = 0.001833m$$

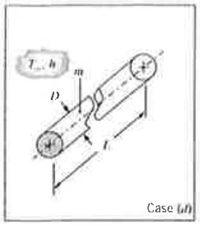
ignoring ends

(L_{c1})

$$B_i = \frac{h(L_{c1})}{k} = \frac{(37\frac{w}{m^2k})(0.001833m)}{177W/mk} = 0.000383 < 0.1$$

Lumped capacitance approx. is valid.

D) An $L = 300\,mm$ long solid stainless steel rod of diameter $D = 13\,mm$ & mass $m = 0.328\,kg$ is exposed to a convection coefficient of $h = 30\,\frac{W}{m^2 k}$



Case (d)

$$L_{c_1} = \frac{V}{A_s} = \frac{\frac{\cancel{\pi} D^{\cancel{2}}}{4} L}{\cancel{2} \times \frac{\cancel{\pi} D^{\cancel{2}}}{\cancel{4}_2} + \cancel{\pi}\cancel{D} L} = \frac{\frac{DL}{4}}{\frac{D}{2} + L} = \frac{\frac{(13mm)(300mm)}{4}}{(\frac{13}{2} + 300)mm} = 0.003181\,m$$

Finding the Max center to Surface L_{c_2} = radius or $= \frac{D}{2} = \frac{0.13mm}{2} = 0.0065m$

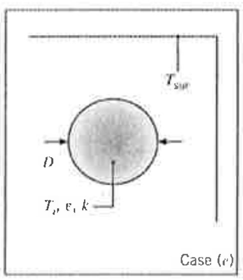
(L_{c_1}) $Bi_1 = \frac{hL_{c_1}}{k} = \frac{(30\frac{W}{m^2k})(L_{c_1})}{13.4\frac{W}{mk}} = 0.007122 < 0.1$ ✓

(L_{c_2}) $Bi_2 = \frac{hL_{c_2}}{k} = \frac{(30\frac{W}{m^2k})(L_{c_2})}{13.4\frac{W}{mk}} = 0.014552 < 0.1$ ✓

Find k by using mass & volume => density

$$\rho = \frac{0.328\,kg}{\frac{\pi D^2}{4} \times L} = 8237.13\,\frac{kg}{m^3} \Rightarrow k = 13.4\,\frac{W}{mk}$$

E) A solid sphere of diameter $D = 13\,mm$ & thermal conductivity $k = 130\,\frac{W}{mk}$ is suspended in a large vacuum oven w/ internal wall temps of $T_{sur} = 18°C$. The initial sphere temp. is $T_i = 100°C$ & $\varepsilon = 0.75$



Case (e)

$$L_{c_1} = \frac{V}{A_s} = \frac{\frac{\cancel{\pi} D^{\cancel{3}}}{6}}{\cancel{\pi D^2}} = \frac{D}{6} = 0.002167m$$ ✓

max center to surface

$$L_{c_2} = \frac{D}{2} = 0.0065m$$ ✓

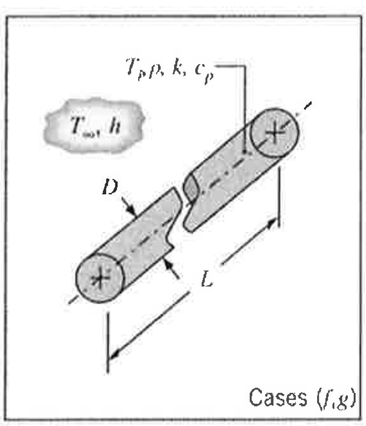
calculate Bi, $h_r = \varepsilon\sigma(T_i + T_{sur})(T_i^2 + T_{sur}^2) = 6.32812\,W/m^2 K$

#1 $Bi = \frac{h_r(0.002167m)}{130\frac{W}{mk}} = 0.000105 < 0.1$ ✓

#2 $Bi = \frac{h_r(0.0065m)}{130\frac{W}{mk}} = 0.000316 < 0.1$ ✓

Lumped capacitance approximation is valid for both definitions

f) A long cylindrical rod of diameter $D = 20\,mm$, density $\rho = 2300\,kg/m^3$, specific heat $c_p = 1750\,J/kg \cdot k$, and thermal conductivity $k = 16\,W/mk$ is suddenly exposed to convective conditions w/ $T_\infty = 20°C$. The rod is initially at a uniform temp. of $T_i = 200°C$ and reaches a spatially averaged temp of $T = 100°C$ @ $t = 225s$.



Cases (f, g)

Find h: $h = \frac{\rho V c}{A_s t} \ln\left(\frac{\theta_i}{\theta}\right) = \frac{\rho\left(\frac{\cancel{\pi} D^{\cancel{2}}}{4} \times \cancel{L}\right)c}{\cancel{\pi}\cancel{D} \times \cancel{L}\; t} \ln\left(\frac{(T_i - T_\infty)}{(T - T_\infty)}\right)$

* $h = 72.5332\,\frac{W}{m^2k}$

Now find $L_{c_1} = \frac{V}{A_c} = \frac{(\frac{\cancel{\pi} D^{\cancel{2}}}{4} \times \cancel{L})}{\cancel{\pi D} \times \cancel{L}} = \frac{D}{4} = 5mm = 0.005m$ ✓

max cent. to surface

@ $L_{c_2} = \frac{D}{2} = 10mm = 0.010m$ ✓

#1 $Bi = \frac{hL_{c_1}}{k} = 0.0227 < 0.1$ ✓

#2 $Bi = \frac{hL_{c_2}}{k} = 0.0453 < 0.1$ ✓

Lumped Capacitance approx is valid for both cases.

G) Repeat part F) w/ Rod diameter of 200 mm.

$$*\ h = \frac{\rho D c}{4t} \ln\left(\frac{(T_i - T_\infty)}{(T - T_\infty)}\right) = 725.332\ \frac{W}{m^2 k}$$

$$L_{c1} = \frac{V}{A_s} = \frac{D}{4} = \frac{200mm}{4} = 50mm = 0.050 m$$

$$L_{c2} = \frac{D}{2} = 0.100 m$$

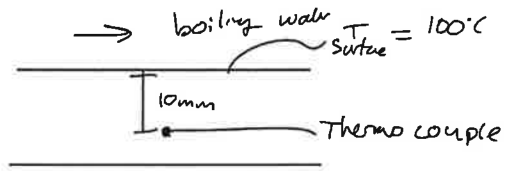
#1 $Bi = \frac{h L_{c1}}{k} = 2.267 > 0.1$ X

#2 $Bi = \frac{h L_{c2}}{k} = 4.533 > 0.1$ X

) lumped capacitance approx is not valid.

PROBLEM 5.70

A procedure for determining the thermal conductivity of a solid material involves embedding a thermocouple in a thick slab of the solid and measuring the response to a prescribed change in temp. at one surface. Consider the arrangement for which the thermocouple is embedded 10mm from a surface that is suddenly brought to a temp of 100°C by exposure to boiling water. If the initial temp of the slab was 30°C and the thermocouple measures a temperature of 65°C, 2 min after the surface is brought to 100°C, what is the thermal conductivity? The density & specific heat of the solid are known to be 2200 kg/m³ & 700 J/kg·k.



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$t = 2 \times 60 = 120 s$
& $T_i = 30°C$
$T = 65°C$

* Assume constant properties
* One Dimension
* time dependence
* Assume semi-infinite solid *
* Assume constant surface temp

$$\frac{T(x,t) - T_s}{T_i - T_s} = erf\left(\frac{x}{2\sqrt{\alpha t}}\right)$$

where $\alpha = \frac{k}{\rho c_p}$

$$\left(\frac{65°C - 100°C}{30°C - 100°C}\right) = erf(w)$$

$$\underbrace{0.5 = erf(w)}_{\text{appendix B}}$$

(using $erf(w) = 0.5$)

w	$erf(w)$
0.44,	0.46622
0.48,	0.50275

$$\frac{x}{0.48} = \frac{0.5}{0.50275} \Rightarrow x = 0.47737$$

$$0.4774 = \frac{x}{2\sqrt{\alpha t}}$$

$$2(0.4774)\sqrt{\alpha t} = x$$

$$2(0.4774)\sqrt{\frac{k}{\rho c_p} t} = x$$

$$\sqrt{\frac{k}{\rho c_p} t} = \frac{0.010 m}{2(0.4774)}$$

$$k = \left(\frac{0.010 m}{2(0.4774)}\right)^2 \frac{\rho c_p}{t}$$

$$k = \left(\frac{0.010}{2(0.4774)}\right)^2 m^2 \times \frac{2200 \frac{kg}{m^3} \times 700 \frac{kJ}{kg \cdot k}}{120 s}$$

$$k = 1.41\ \frac{W}{mk}$$

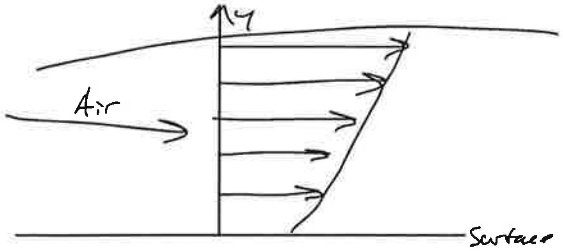
$$\boxed{\text{Thermal Conductivity } k = 1.41\ W/mk}$$

PROBLEM 6.3

In a particular application involving airflow over a surface, the boundary layer temp. distribution may be approximated as

$$\frac{T-T_s}{T_\infty-T_s} = 1-\exp\left(-Pr\frac{u_\infty y}{\nu}\right)$$

where y is the distance normal to the surface and the Prandtl number, $Pr = c_p\mu/k = 0.7$, is a dimensionless fluid property. If $T_\infty = 400k$, $T_s = 300k$ and $u_\infty/\nu = 5000\ m^{-1}$, what is the surface heat flux?



$$\frac{T-T_s}{T_\infty-T_s} = 1-\exp\left(-Pr\frac{u_\infty y}{\nu}\right)$$

$$T(y) = T_s + (T_\infty - T_s)\left(1-\exp\left(-Pr\frac{u_\infty y}{\nu}\right)\right)$$

We want to find

$$\frac{\partial T}{\partial y} = \cancel{/} + (T_\infty - T_s)\left(\cancel{-}\left(\cancel{-}Pr\frac{u_\infty}{\nu}\right)\exp\left(-Pr\frac{u_\infty y}{\nu}\right)\right)$$

25/25

$$q''_{surface} = -k\left.\frac{\partial T}{\partial y}\right)_{y=0} = \qquad \frac{\partial T}{\partial y} = (T_\infty - T_s)\left(Pr\frac{u_\infty}{\nu}\right)\exp\left(-Pr\frac{u_\infty y}{\nu}\right)$$

$$q''_{surface} = -k(T_\infty - T_s)\left(Pr\frac{u_\infty}{\nu}\right)\cancel{\exp\left(-Pr\frac{u_\infty y}{\nu}\right)}\Big]_{y=0}$$

$$q''_{surface} = -\underset{air}{k}\,(T_\infty - T_s)\left(Pr\frac{u_\infty}{\nu}\right) \quad e^0 = 1$$

$$q''_{surface} = -\left(0.0263\ \frac{W}{m\cancel{k}}\right)(400-300)\cancel{k}\times 0.7\times 5000\,\frac{1}{m}$$

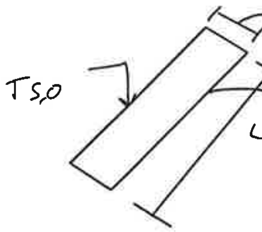
$$\boxed{q''_{surface} = -9205\ \frac{W}{m^2}}$$ heat into the surface

PROBLEM 6.34

The defroster of an automobile functions by discharging warm air on the inner surface of the windshield. To prevent condensation of water vapor on the surface, the temp of the air & the surface convection coeff. $(T_{\infty,i}, \bar{h}_i)$ must be large enough to maintain the surface temp $T_{s,i}$ that is at least as high as the dewpoint $(T_{s,i} \geq T_{dp})$.

Consider a windshield of length $L = 800mm$ & thickness $t = 6mm$ and driving conditions for which the veheicle moves at a velocity of $V = 70mph$ in ambient air @ $T_{\infty,o} = -15°C$. From labratory experiments preformed on a model of the vehicle, the avg convection coeff. on the outer surface of the windshield is known to be correlated by an expression of the form $\overline{Nu}_L = 0.030 Re_L^{0.8} Pr^{1/3}$, where $Re_L = VL/\nu$. Air properties may be approximated as $k = 0.023\ W/mk$, $\nu = 12.5\times10^{-6}\ m^2/s$, & $Pr = 0.71$. $T_{dp} = 10°C$ & $T_{\infty,i} = 50°C$ what is the smallest value of $\bar{h}_i$ required to prevent condensation on the inner surface?

* $(T_\infty, \bar{h}_i)$ large enough to maintain a surface temp $T_{s,i}$ @ dewpoint $(T_{s,i} \geq T_{dp})$



$L = 800mm$
$t = 6mm$
$V = 70\ mph$
Air → $T_{\infty,o} = -15°C$
$\overline{Nu}_L = 0.030\ Re_L^{0.8}\ Pr^{1/3}$
$Re_L \equiv VL/\nu$
Air → $k = 0.023\ W/mK$
$\nu = 12.5 \times 10^{-6}\ m^2/s$
$Pr = 0.71$
$T_{dp} = 10°C$
$T_{\infty,i} = 50°C$

* assume constant prop.
* Steady State
* One Dimensional

FIND smallest value of $\bar{h}_i$ req. to prevent condensation on the inner surface?

Heat flux travels out of the windshield

∴

Resistance is in series

FROM BOOK
Glass: $k = 1.4 \frac{W}{mK}$

$$\frac{T_{\text{inside car}} - T_{\text{inside car surface}}}{R_{\text{convection}}} = \frac{T_{\text{inside car surface}} - T_{\text{outside}}}{R_{cond} + R_{convec}}$$

⇓ correct symbols

(25/25)

$$\frac{T_{\infty,i} - T_{s,i}}{\frac{1}{\bar{h}_i}} = \frac{T_{s,i} - T_{\infty,o}}{\frac{t}{k_{glass}} + \frac{1}{\bar{h}_o}}$$

thickness "L"

$$\overline{Nu}_L = \frac{\bar{h}_o L}{k} = 0.030\ Re_L^{0.8}\ Pr^{1/3} \Rightarrow \bar{h}_o = \frac{k}{L}\left(0.030\ Re_L^{0.8}\ Pr^{1/3}\right)$$

air; length?

0.71

$$\bar{h}_o = \frac{0.023 \frac{W}{mK}}{0.800m}\ 0.030\ Re_L^{0.8}\ Pr^{1/3}$$

$$\bar{h}_o = 84.6208\ \frac{W}{m^2K}$$

Finding Reynolds #

$$Re_L = \frac{VL}{\nu} = \left(70mph \times \frac{1609.34\ meters}{1mile} \times \frac{1Hour}{3600seconds}\right) \frac{0.800m}{12.5 \times 10^{-6} \frac{m^2}{s}} = 2.00273 \times 10^6$$

velocity

$T_{s,i} = T_{dp}$ *

$$\bar{h}_i = \frac{(T_{s,i} - T_{\infty,o})}{(T_{\infty,i} - T_{s,i})}\left(\frac{1}{\frac{t}{k_{glass}} + \frac{1}{\bar{h}_o}}\right)$$

$$\bar{h}_i = \frac{(10 - (-15))\cancel{K}}{(50 - 10)\cancel{K}}\left(\frac{1}{\frac{0.006m}{1.4 \frac{W}{mK}} + \frac{1}{84.6208 \frac{W}{m^2K}}}\right) = 38.81\ \frac{W}{m^2K}$$

$$\boxed{\bar{h}_i = 38.81\ \frac{W}{m^2K}}$$